PHYC10005 Physics 1: Fundamentals –Lecture 5

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Physics, Room 501

(Please switch off your mobile phone.)

Summary of Lecture 4

The net force is the vector sum of all the forces acting on an object.

The net force causes a body to accelerate with magnitude inversely proportional to its mass

effect

cause

$$\vec{a} = \frac{\vec{F}_{net}}{m}$$

Newton's second law of motion

$$\vec{w} = (mg, \text{downwards})$$

Newton II - G 4.4

effect

cause

$$\vec{a} = \frac{\vec{F}_{net}}{m}$$

Newton's second law of motion

SI unit of force:

kg

m s^{-2}

Newton (N)

$$\vec{F} = m\,\vec{a}$$

$$1\ \mathrm{N} \equiv 1\ \frac{\mathrm{kg\ m}}{\mathrm{s}^2}$$

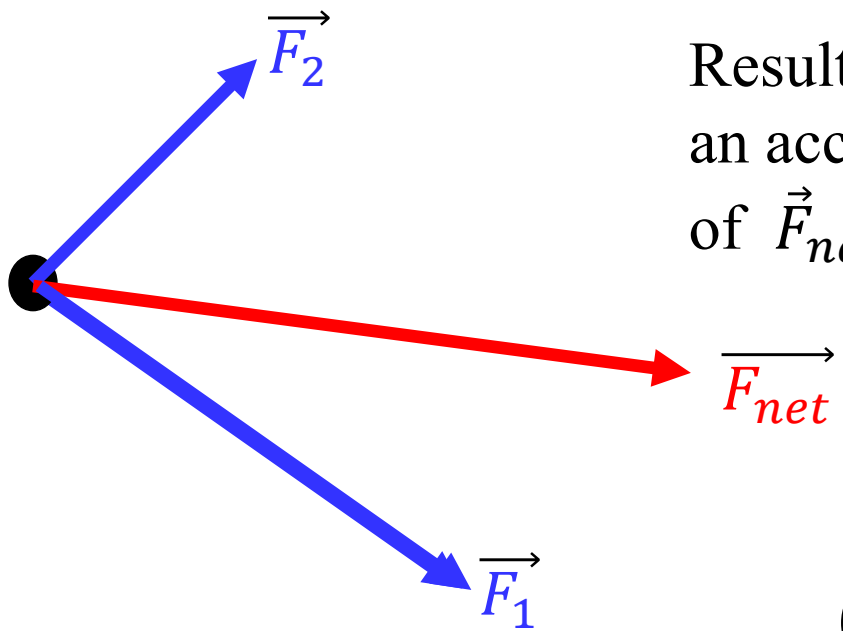
Force is a Vector!

effect

cause

$$\vec{a} = \frac{\vec{F}}{m}$$

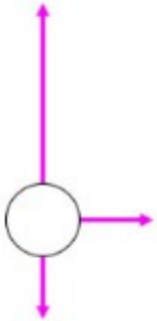
Newton's second law of motion



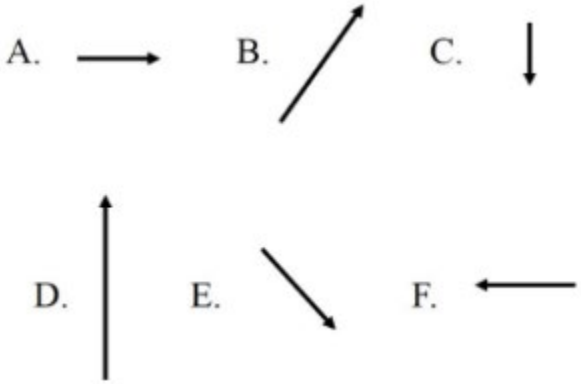
Result of $\vec{F}_1 and\ \vec{F}_2$ is an acceleration in direction of $\vec{F}_{net}$

D Vector addition of forces

Forces as Vectors



Three forces act on an object. In which direction does the object accelerate?



A

B

C

D

E

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Newton III - G4.5

‘For every action, there is an equal and opposite reaction’

Important points about the language of this law:

- Newton is talking about an *interaction* between two objects.
- Newton III is specifically about *force*, and there is no more general meaning to the word ‘action’ in this context.
- Nothing in Newton III specifies which force is the ‘action’ and which is the ‘reaction’. We are free to label them either way.

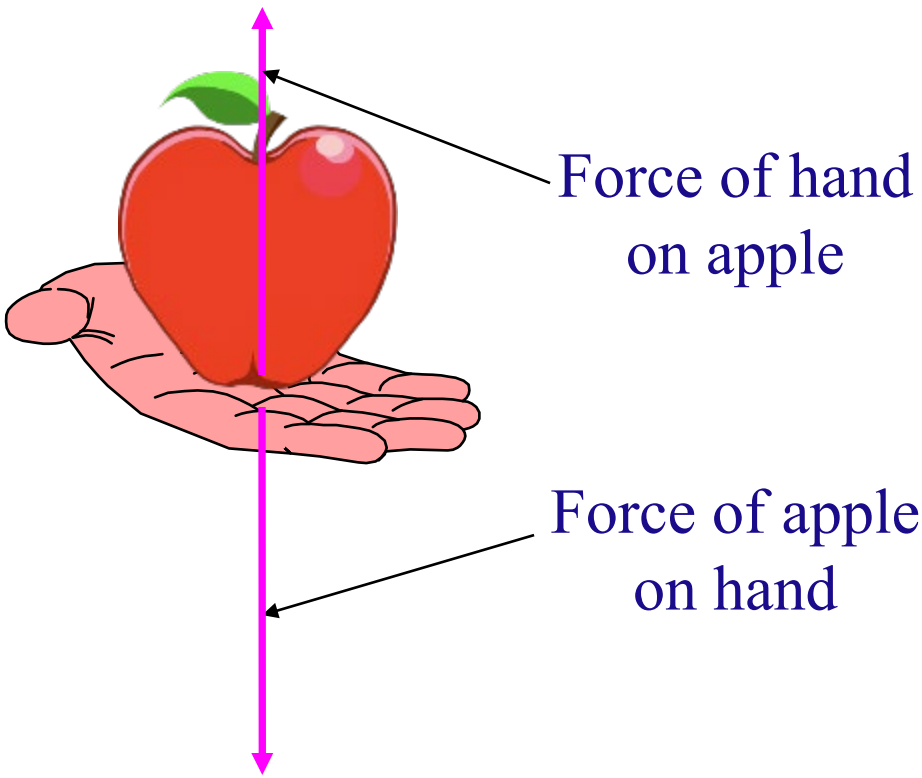
Vital points in understanding Newton III:

- Forces in an action/reaction pair *always* act on *different* objects.
- Action/reaction forces always have equal magnitudes and opposite directions.

D Jet propulsion

Newton III – Action/reaction force pairs - G4.5

All forces result from interactions between objects:

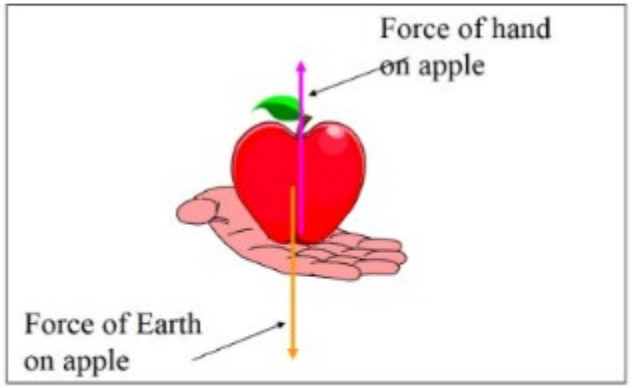


These two forces have equal magnitude but opposite directions, and are known as an 'action/reaction pair' or an 'interaction pair'.

Action-reaction (air track)

Apple on hand

Consider all the forces on the apple.



Forces are equal and opposite. Is this an action/reaction pair?

Yes

No

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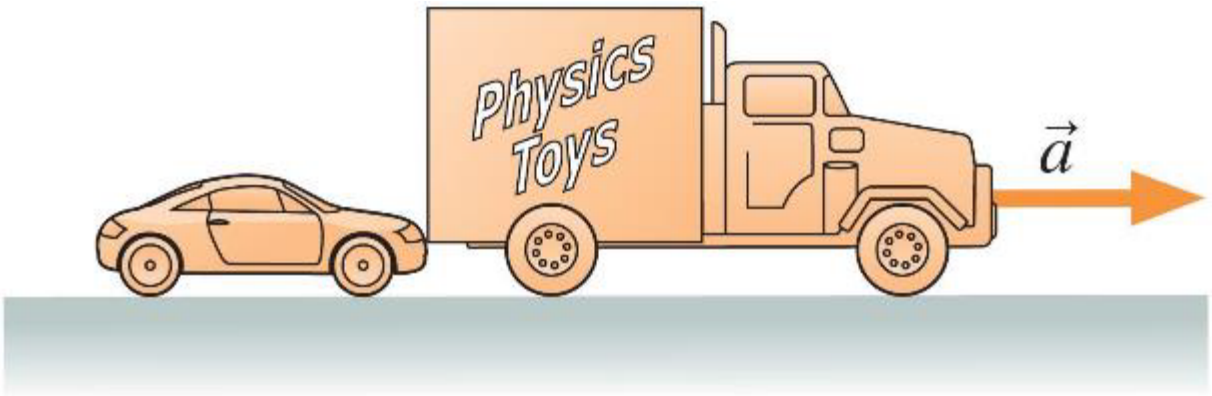
Question



A small car is pushing a large truck that has a dead battery. The mass of the truck is much greater than the mass of the car. Which of the following is true?

A. The car exerts a force on the truck, but the truck exerts no force on the car.

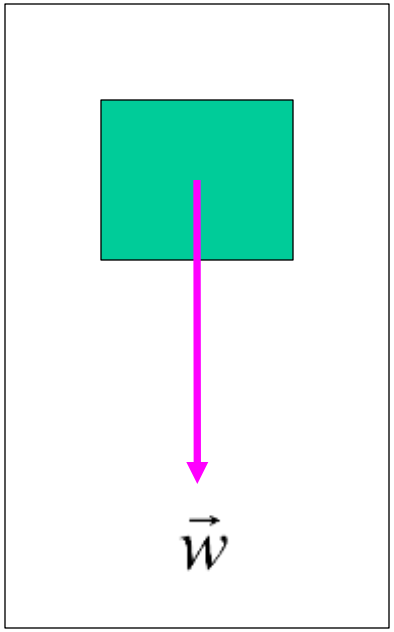
B. The car exerts a larger force on the truck than the truck exerts on the car.

C. The car exerts the same amount of force on the truck as the truck exerts on the car.

D. The truck exerts a larger force on the car than the car exerts on the truck.

E. The car exerts no force on the truck; the truck simply gets in the way.

Weight – G4.6



- We have already seen that all objects in free fall accelerate at the same rate near the Earth's surface.

$$\vec{a} = (g, \text{downwards})$$

- According to Newton I and Newton II, an object in free fall must therefore have a net force on it:

$$\vec{F} = m\vec{a} = \vec{w} = (mg, \text{downwards})$$

- This force is called the *weight force.* Its magnitude is the *weight* of the object.

$$w = mg$$

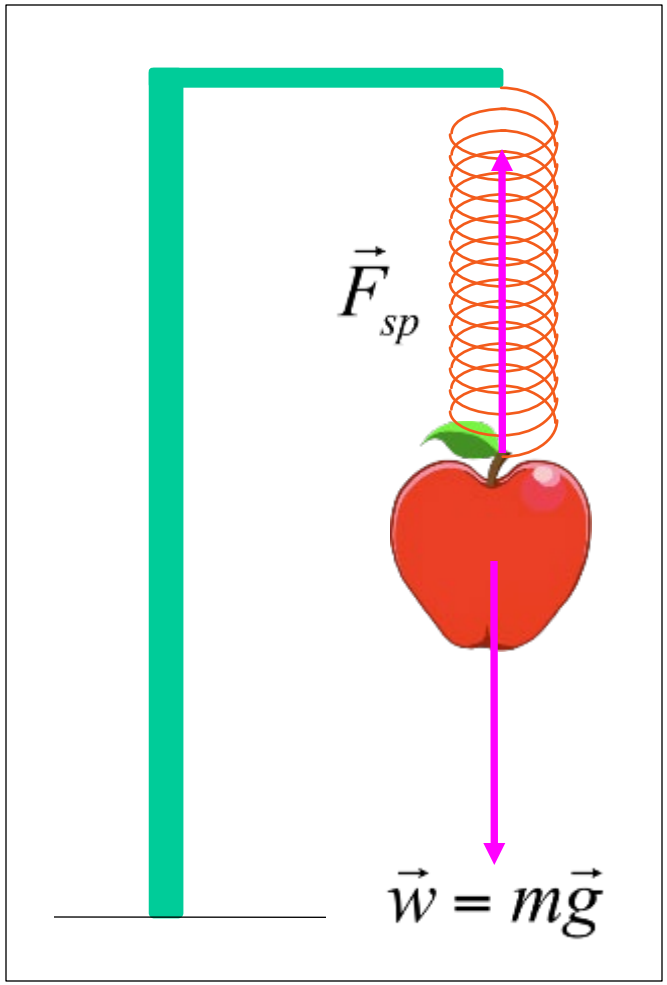
- Even though the above derivation is for free fall, there is *always* a weight force on an object, even when it is not accelerating (as long as the local value of g is not zero).

Mass vs weight - G 4.3,4.6

$$\vec{w} = (mg, \text{downwards})$$

Mass	Weight
⌖ A measure of the *inertia* of an object (resistance to acceleration).	⌖ The magnitude of the gravitational *force* exerted on an object.
⌖ Mass is a *scalar* quantity.	⌖ Weight force is a *vector*.
⌖ Depends only only the 'amount of substance' in an object – i.e. mass is an intrinsic property.	⌖ Depends on both the mass *and* the local acceleration due to gravity, *g*.

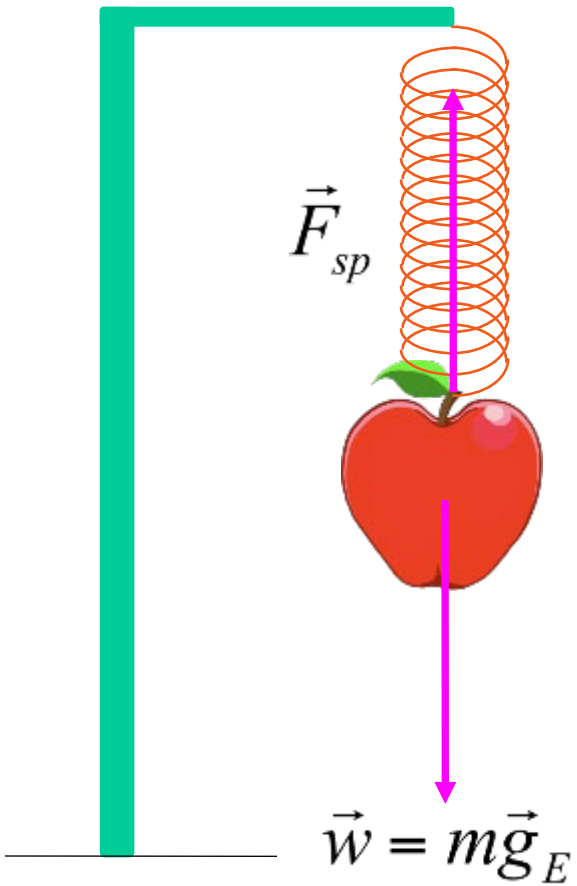
Measuring weight – G4.6



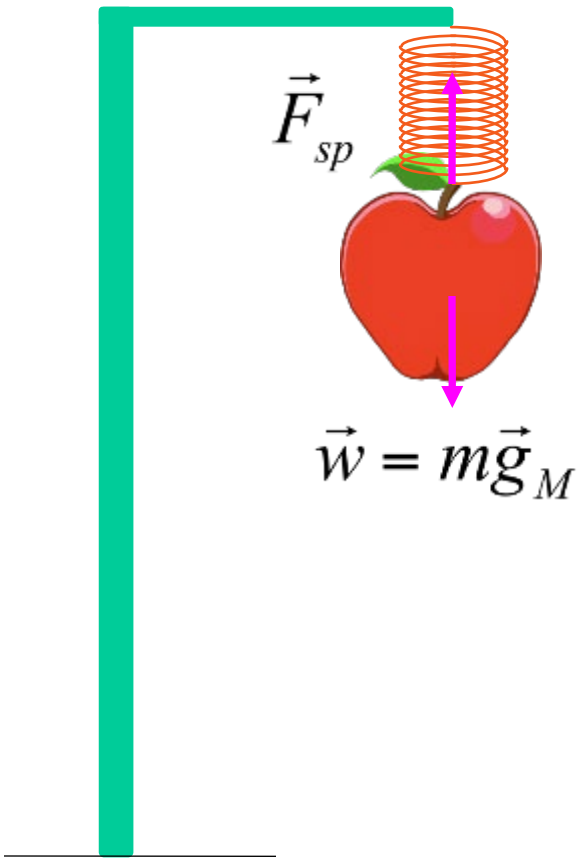
Spring balance

- If the apple is not accelerating, net force on it is zero. (Newton I)
- Therefore, spring force = weight force.
- Measure extension of the spring to determine spring force, and hence the weight.
  - We will look at springs in more detail in a moment…

Weight depends on local gravity – G4.6



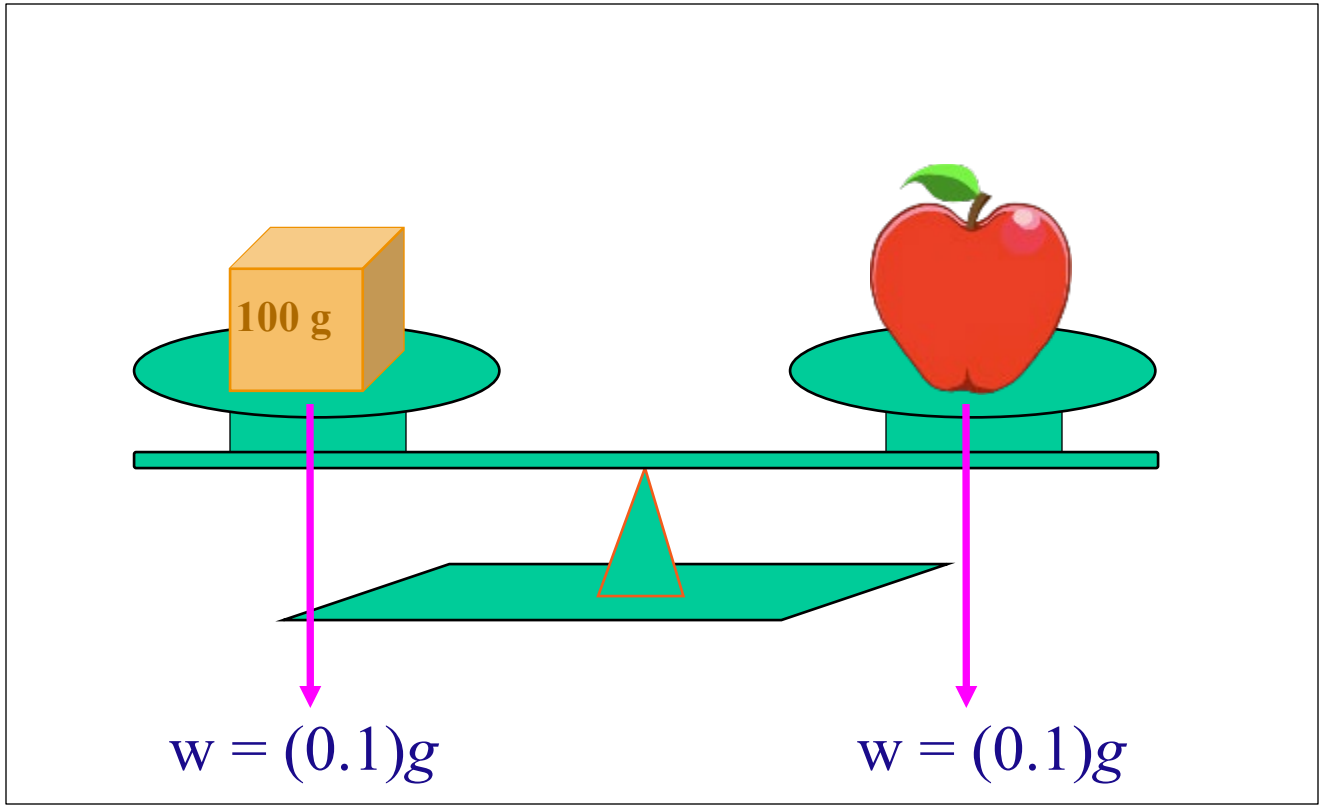
On Earth (g_E = 9.8 m/s^2)



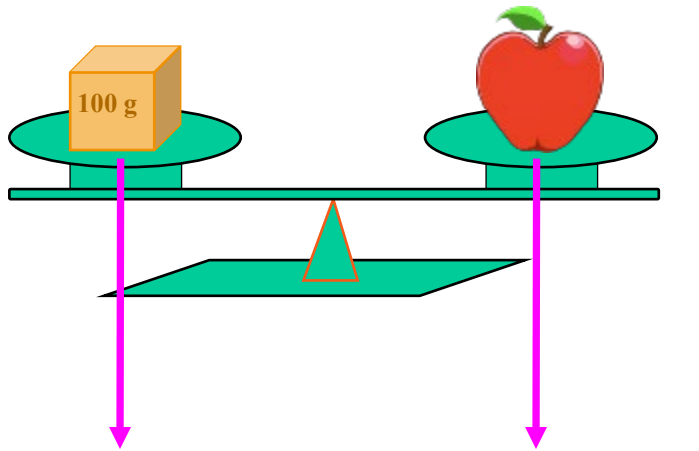
On Mars (g_M = 3.8 m/s^2)

Measuring mass - G 4.3,4.6

- Apply known force to object and measure its acceleration (as in last lecture); OR
- Compare unknown mass to a known mass using a *pan balance*.

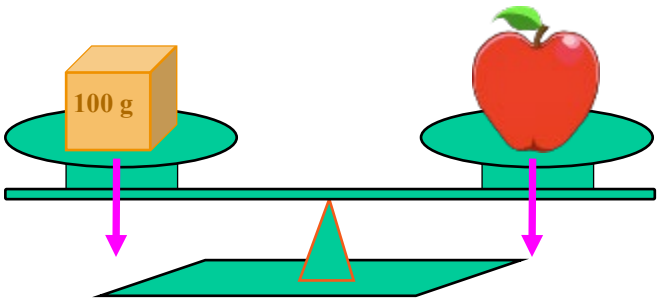


Mass is intrinsic



$w = (0.1)g_E$ $\quad$ $w = (0.1)g_E$

On Earth ($g_E = 9.8$ m/s^2)

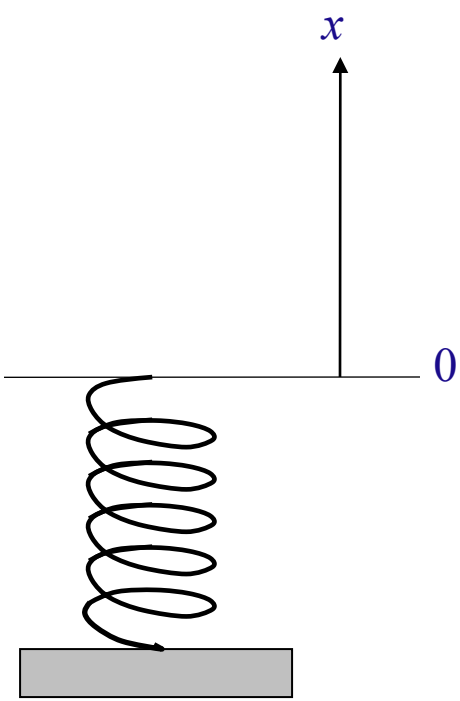


$w = (0.1)g_M$ $\quad$ $w = (0.1)g_M$

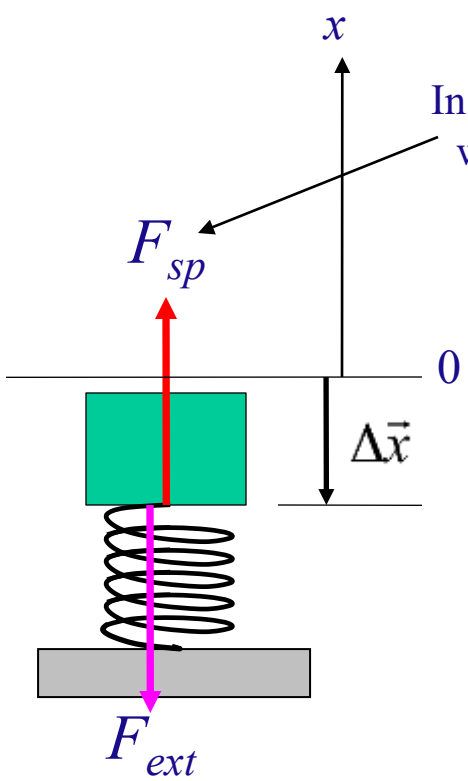
On Mars ($g_M = 3.8$ m/s^2)

Spring forces - G 6.4

Spring at its natural length (unstretched)



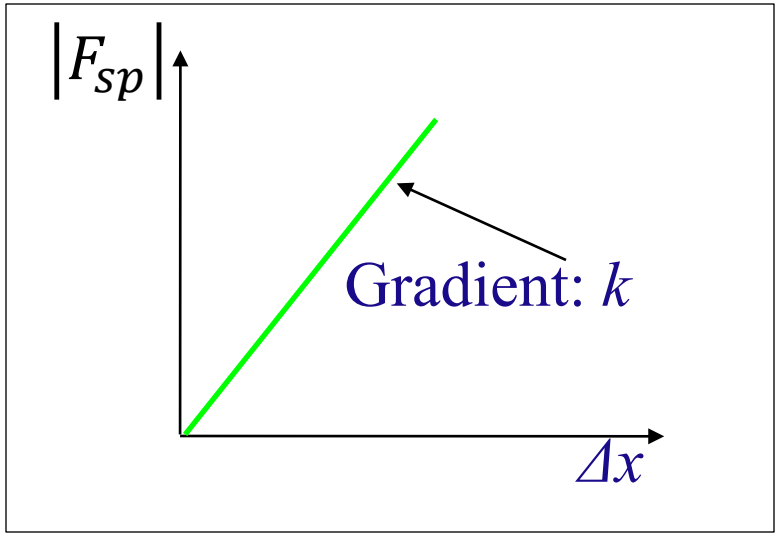
Apply a known force F_{ext} to the spring (e.g. place an object with a known weight on it.) Measure the compression of the spring, Δx.



In equilibrium, spring pushes back with a force equal to the weight (Newton I and III).

Spring force is in opposite direction to $\Delta\vec{x}$

Hookes' law - G6.4



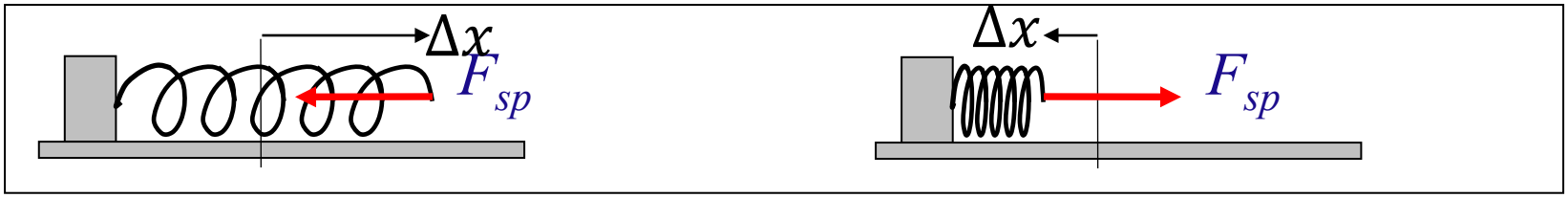
Result: spring force is proportional to extension of spring.

$$|F_{sp}| \propto \Delta x$$

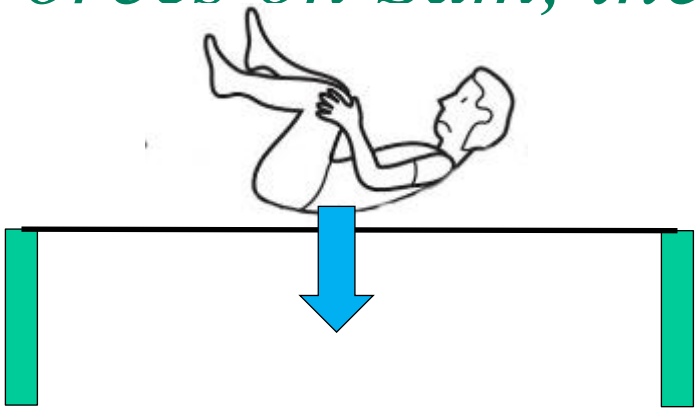
or

$$\vec{F}_{sp} = -k\Delta\vec{x}$$ Hookes' law

- Constant of proportionality k is known as the *spring constant*.
- k is a characteristic property of a particular spring – different for different springs.
- Minus sign indicates that the spring force is always in the opposite direction to the extension or compression of the spring.

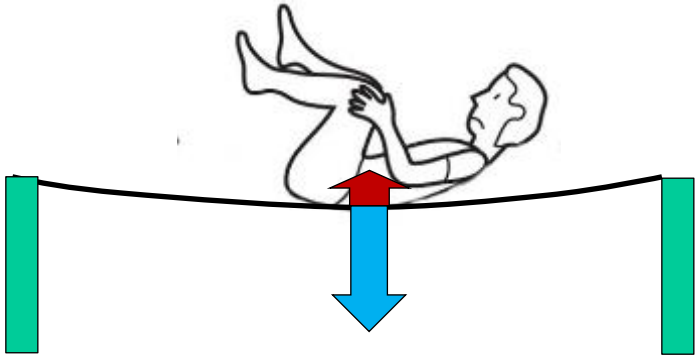


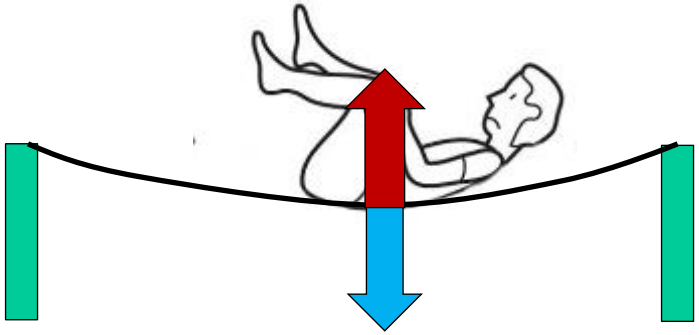
Forces on Sam, the trampoliner



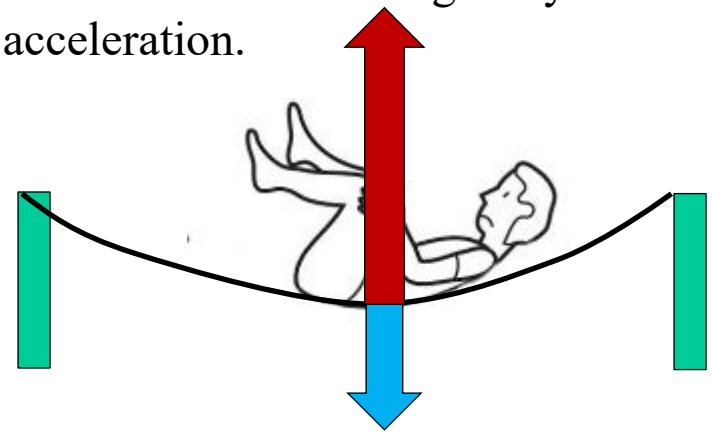
Sam has not hit trampoline. Only force is his weight

Sam touches trampoline and starts to stretch the trampoline springs. Acceleration down decreases with the opposing force to weight



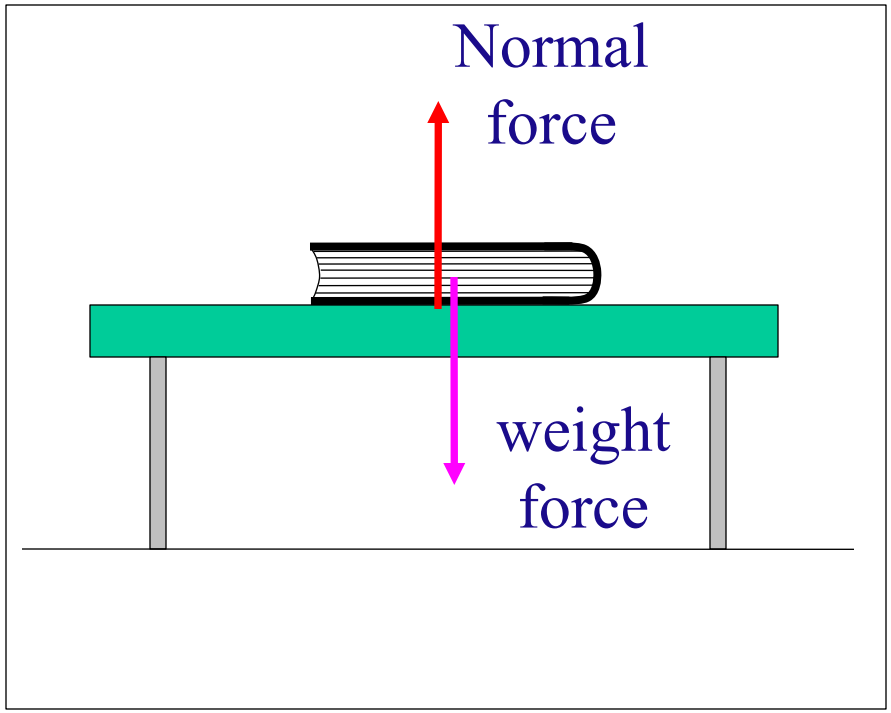


Trampoline springs stretch enough to provide enough force to counter that of gravity. Net zero acceleration.



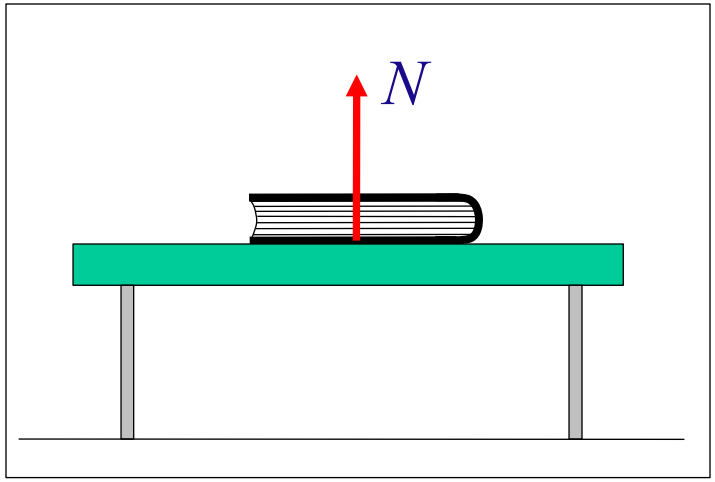
Sam has zero velocity. Trampoline spring forces substantially larger than gravity. Maximum upwards acceleration.

Normal force – G 4.6

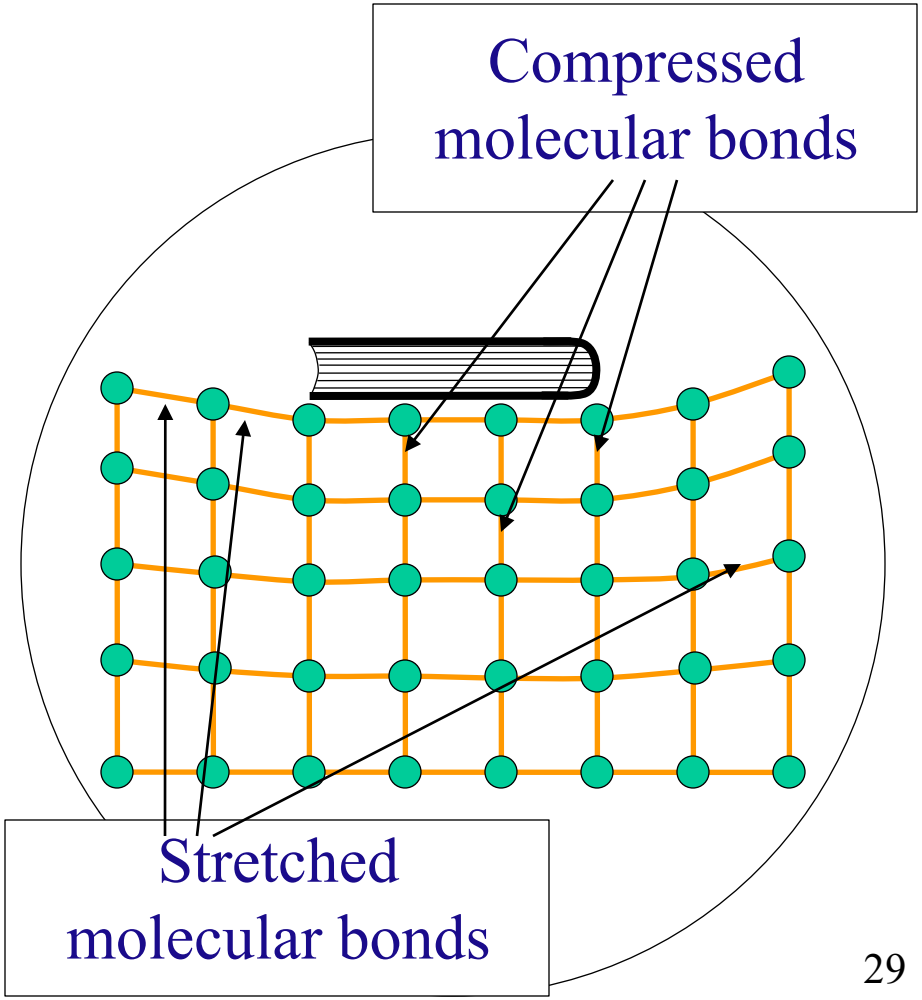


- Consider a book that is stationary on a table.
  - Net force on book must be zero. (Newton I).
  - There is a weight force on the book.
  - Therefore, there must by another force that supports the book.
- The contact force exerted by the table is called the *Normal force*.

Origin of the Normal force

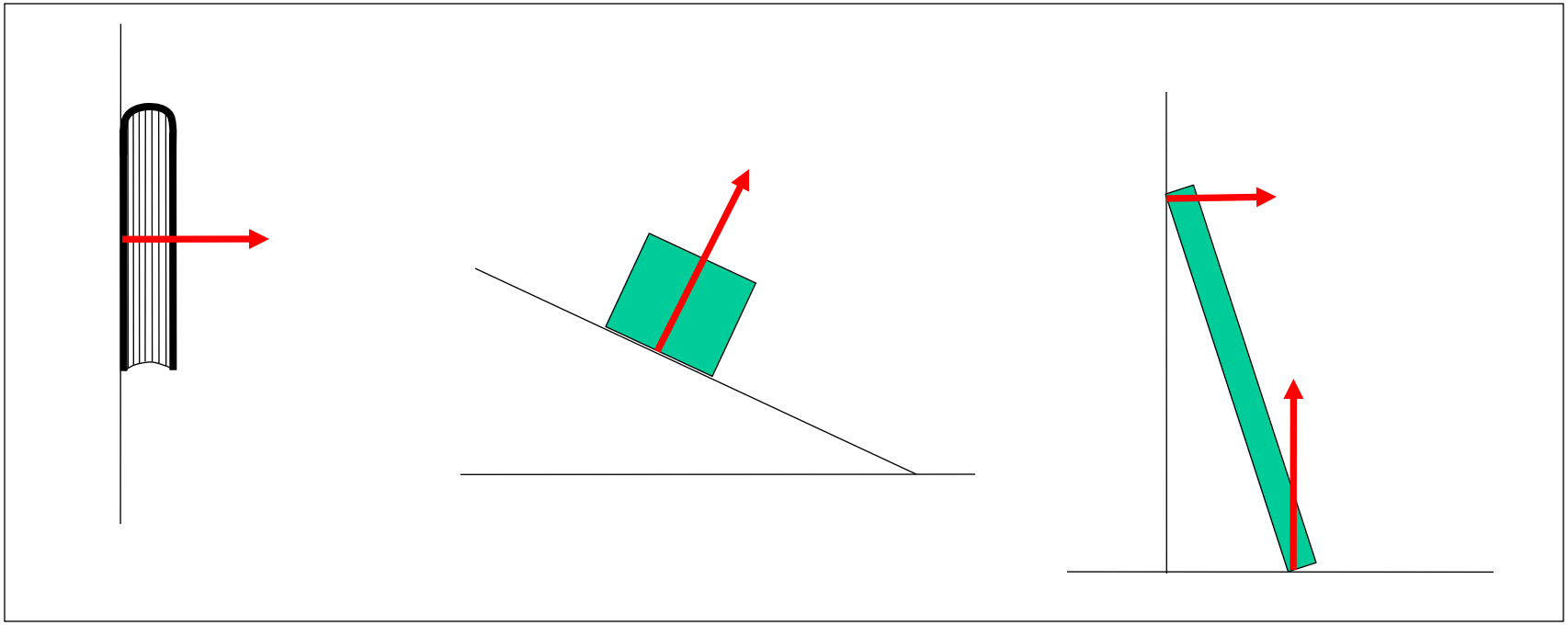


- Molecular bonds in the table act like springs.
- Compression and extension of bonds caused by the book pressing down results in reaction forces pushing upwards (similar to macroscopic spring forces).
- The normal force is the combined force from many molecular bonds.

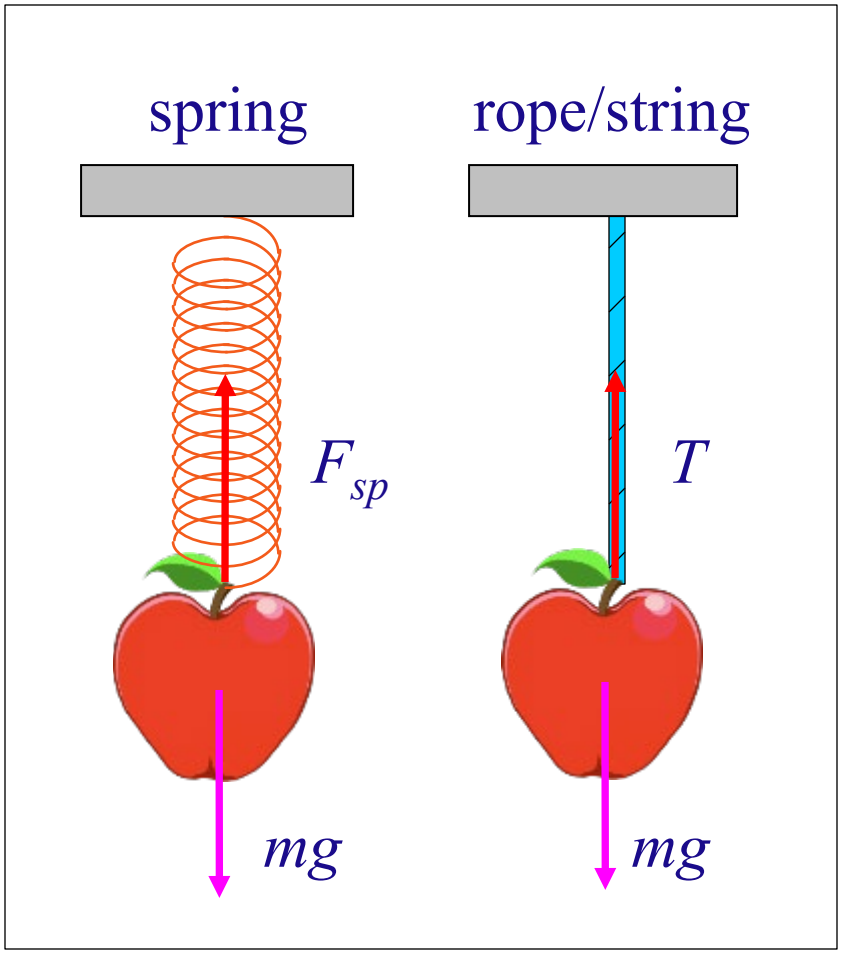


Why 'normal'? - G 4.6

- The Normal force acting on an object pressing against a surface always acts in a direction perpendicular, or '*normal*', to the surface.
  - 'Normal' in this context only means 'perpendicular', not 'ordinary' or 'usual'. There is no such thing as an 'abnormal' force.
- Other examples of the Normal force:



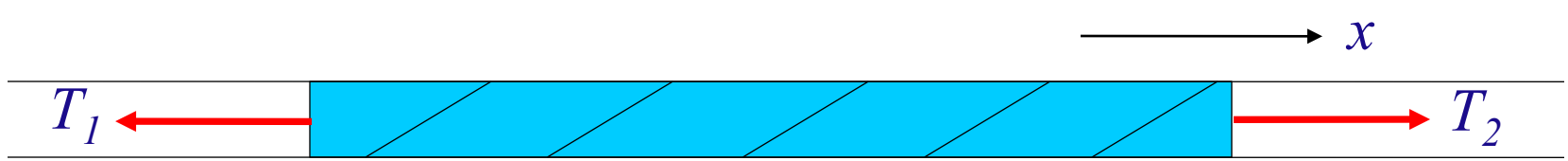
Tension forces - G4.7



- Molecular bonds in a rope or string act like tiny springs.
- The tension force, *T*, is due to the combined effect of the molecular 'spring' forces.
- The effective spring constant *k* is a lot larger for a rope than for a spring, so the rope doesn't stretch very much, even for large forces!
  - In problems, we usually assume that the rope doesn't stretch *at all*. This is the 'inextensible string' approximation.

Tension forces - G 4.7

- Consider the forces on one section of a rope.
  - Mass of section = m



Net force = $T_2 - T_1 = ma$

- If the rope is not accelerating, then $a = 0$. Therefore, $T_1 = T_2$.
- If the rope *is* accelerating, then the difference in tension between the two ends of the section depends on the mass of the section.
- Usually, we assume that the mass of the rope is negligible, especially when the objects attached to it are much more massive than the rope itself. In that case:

Tension is equal throughout a rope/string.

- This is the 'massless string' approximation (usually used in combination with the inextensible string approximation).

D Action-reaction (air track)

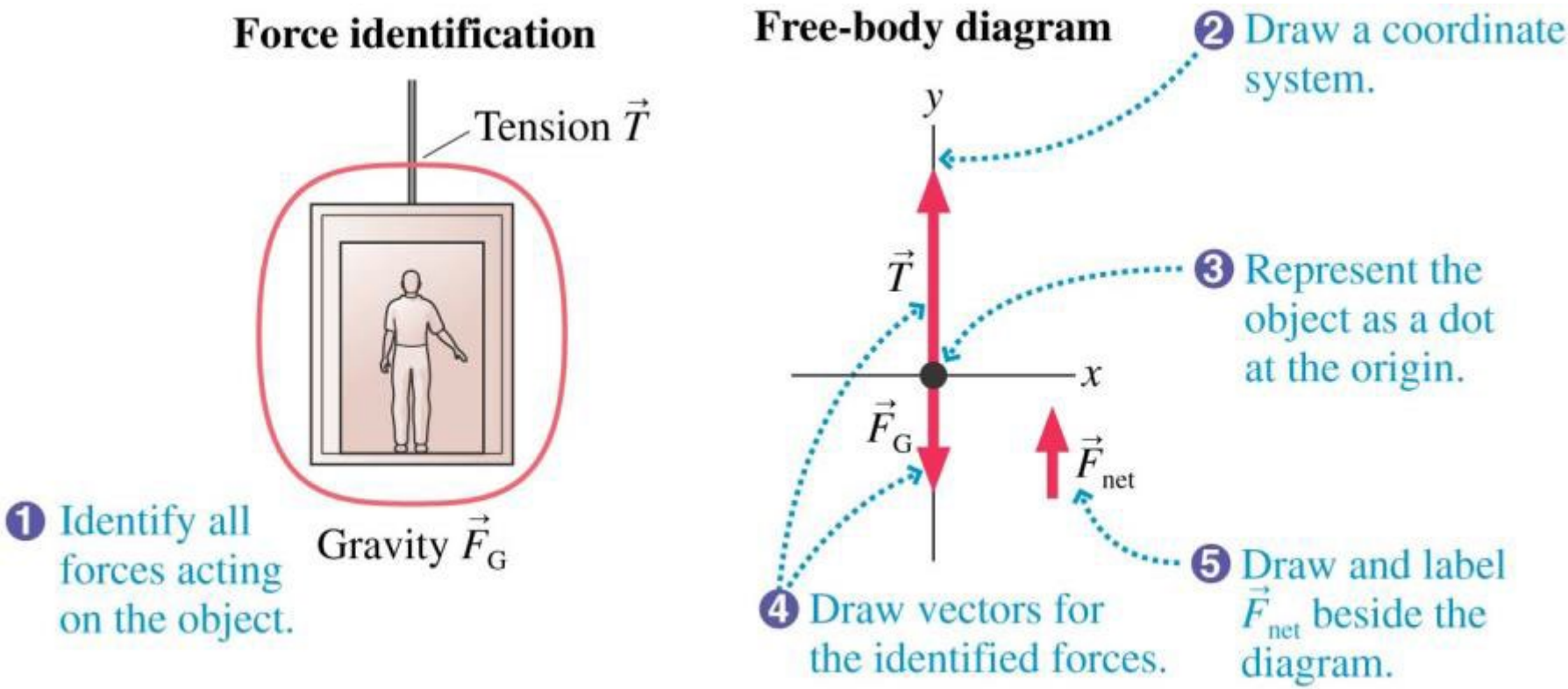
Using Newton's laws to solve problems - G 4.7

Outline of a general strategy for solving force and motion problems

1. Identify the system of interest (which may include several objects).
2. Draw a diagram of the system of interest and its environment.
   - Include every object in the environment that is in direct contact with the system (surfaces, ropes, springs, people…)
3. Locate all points at the boundary of the system where the environment exerts contact forces on the system.
4. Draw and label force vectors for each contact force on the system.
5. Add additional vectors for any long-range forces on the system.
6. Choose a convenient coordinate system and add coordinate axes to your diagram.
7. Apply Newton's laws in each coordinate direction.
   - Find components of the forces and consider the net force on the system.
8. Solve the resulting equations for any unknown forces in the problem, and/or use kinematics to determine the displacement, velocity, etc. of the system.

Identify forces - G4.7

Person in upward <u>accelerating</u> lift



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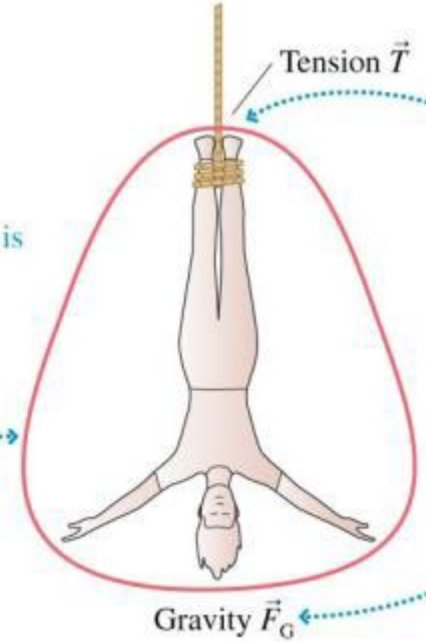
Note *upward force T is larger than downward force* $\vec{F}_g$

Identify forces - G 4.7



1. Identify the object of interest. Here the object is the bungee jumper.
2. Draw a picture of the situation.
3. Draw a closed curve around the object.
4. Locate the points where other objects touch the object of interest. Here the only point of contact is where the cord attaches to her ankles.
5. Name and label each contact force. The force exerted by the cord is a tension force.
6. Name and label long-range forces. Gravity is the only one.

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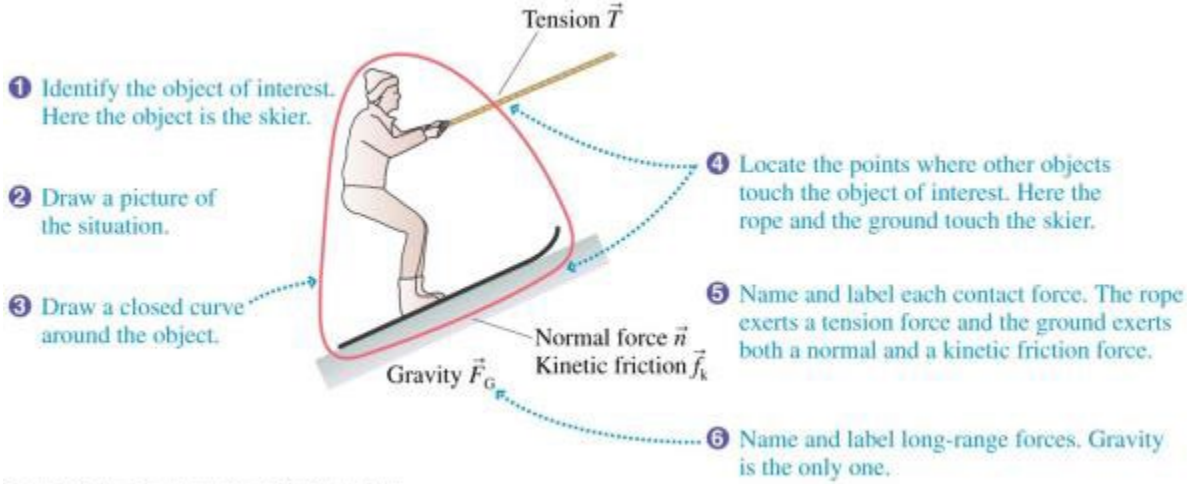
Identify forces - G4.7



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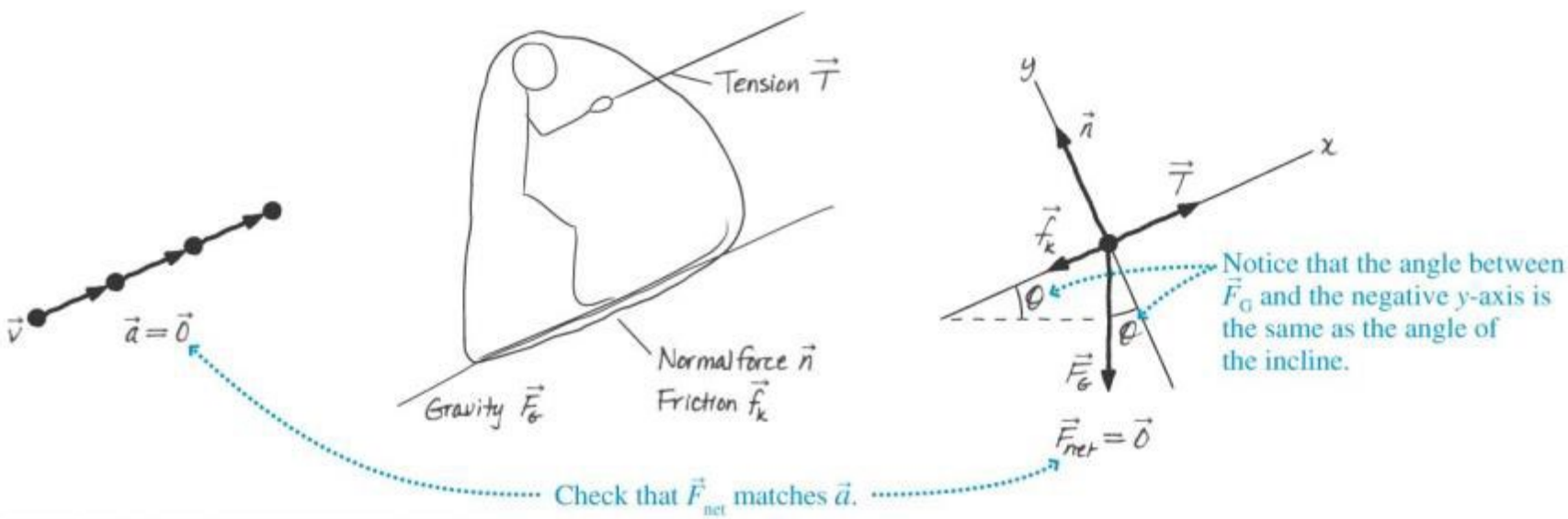
Motion diagram **Force identification** **Free-body diagram**



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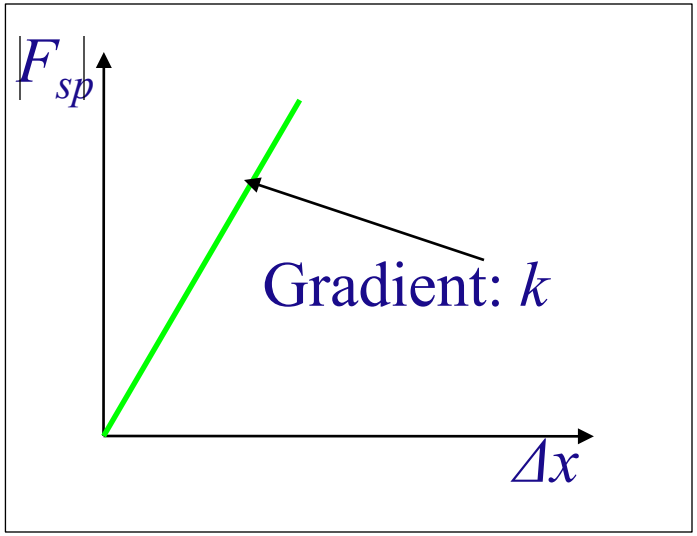
Summary of Lecture 5

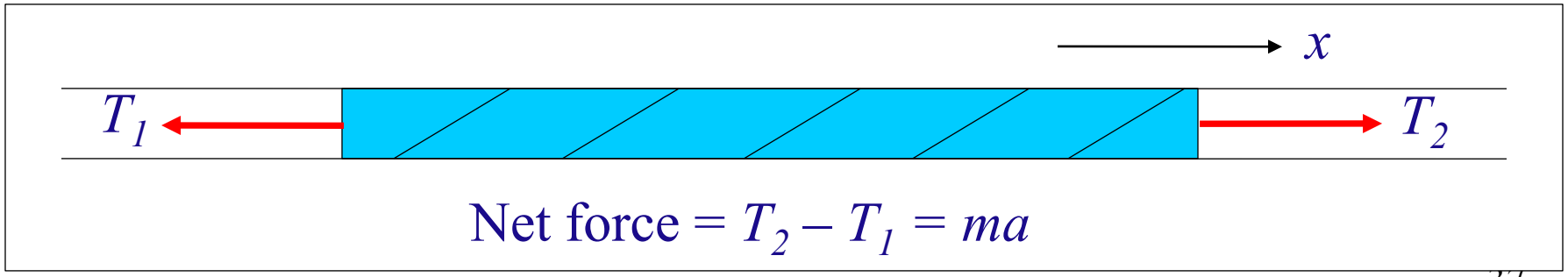


Result: spring force is proportional to extension of spring.

$$F_{sp} \propto \Delta x$$

or

$$\vec{F}_{sp} = -k\Delta\vec{x}$$ Hookes' law

Spring constant

Tension is equal throughout a rope or string